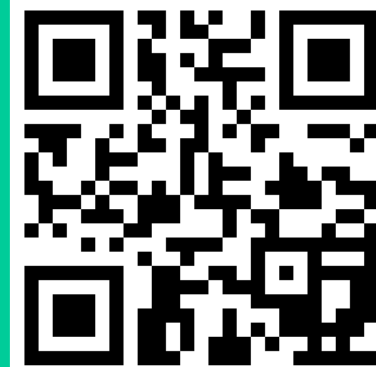




World's widest
range of Post-
Harvest solutions in
Cleaning, Grading,
Sorting, treatment,
Handling & Storage

GRAIN STORAGE SILOS- DESCRIPTION



Our Collaboration For Grain storage Silos



MACHINE BUILDING PLANT "LUBNYMASH"

Was founded in 1888

From 1888 our enterprise was quickly developed and improved its production potential. We've collected huge experience in machine building sphere, that allowed us to reach high professional results and establish production. Equipment of LUBNYMASH trade mark is certified and has all necessary licenses. It's successfully used not only in Ukraine, but in Russia, Belarus, Moldova, Kazakhstan and Uzbekistan.





SILO FUNCTIONS

Grain storage silos are intended for following functions:

- convenient grain reception;
- amount control
- holding of optimum temperature regime of the whole amount of storage components;
- cooling of silo content to prolong its storage term;
- grain discharge in shortest period.

Silos of two types are produced (with conical and flat bottom) their volume can vary from 10,6 cub.m. to 22 171 cub.m.



Effective area usage

Steel silos require less land plot, that is very important in case of area lack (for example in case of object reconstruction) or if the price of land is very high (for example in ports).

SILO ROOF

Roof of the silo notwithstanding the model has 30° inclination, providing the optimum strength. Angle equal 30° degrees correspond to grain slope while filling, that allows to prevent silo overfilling with further roof break down. The geometry of joint locks on roof segments gives an opportunity to enhance isolation and durability of the roof. Acceptable loading for such roof construction is not less than 180 kg/m². Air ducts are placed upon the whole roof area, optionally (on clients request) roof fan can be added.





WALL PANELS

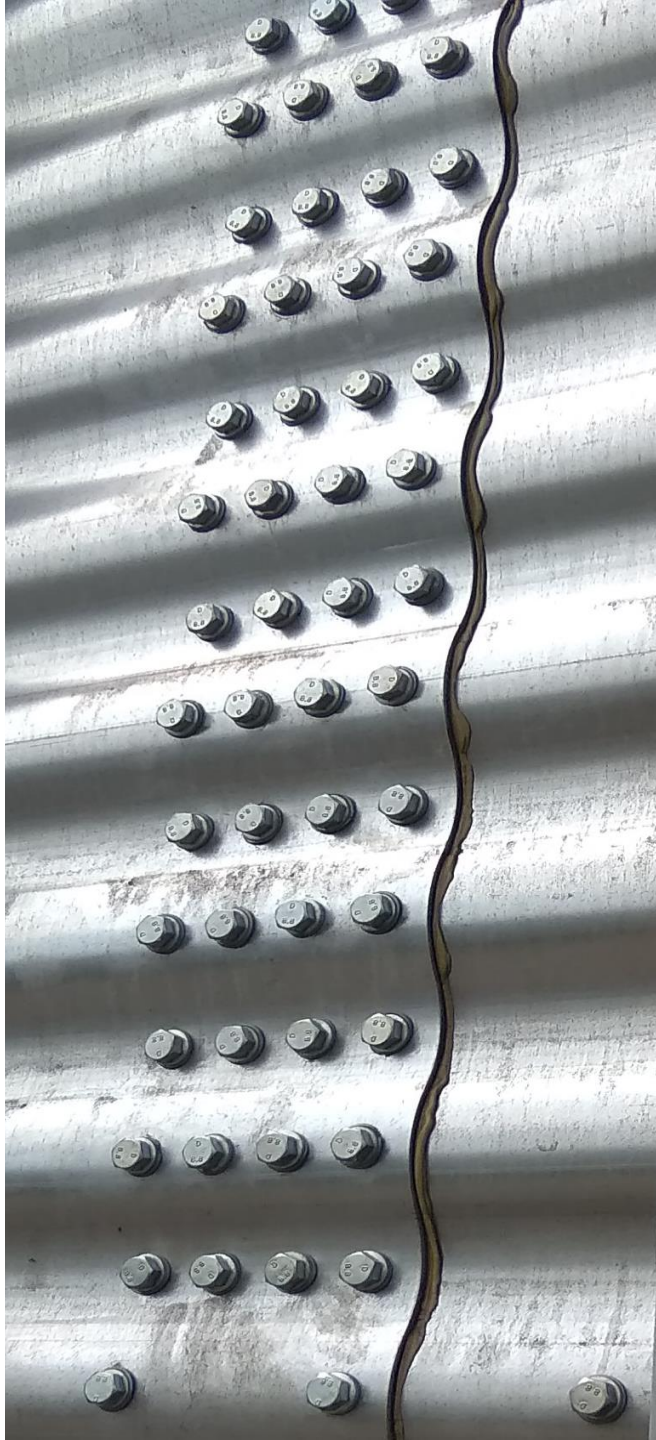
LUBNYMASH silo wall panels are produced on automatic rolling line from galvanized steel coils. The wave pitch is 64 mm with useful sheet height 1152 mm. Steel quality: S 350 GD according to UNE-EN-10346: 2009.

Thickness of galvanization : от 350 до 600 gr/m² according to UNE-EN-10346: 2009. Thickness of steel sheet: from 1 mm to 3 mm depending on layer height.

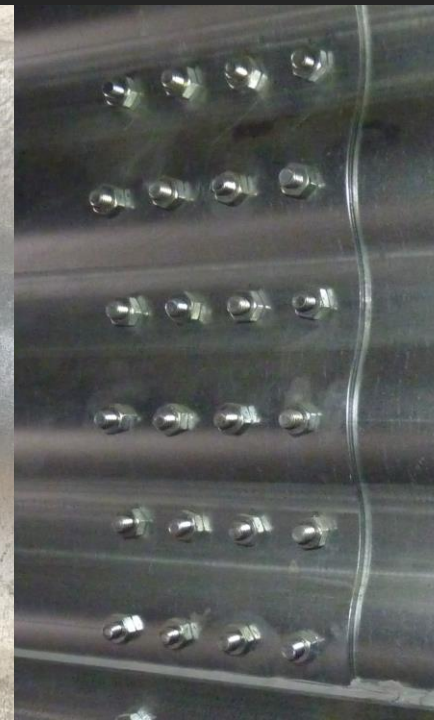
STRENGTHEN RIBS

Strengthen ribs carrying vertical silo loading on foundation are placed outside the silo body. Ribs are produced from strong galvanized steel (galvanization – up to 600 gr/m²), ensuring durability and long life of the construction. Thickness from 1,0 mm to 6,0 mm, omega-shaped profile.

BOLT JUNCTIONS



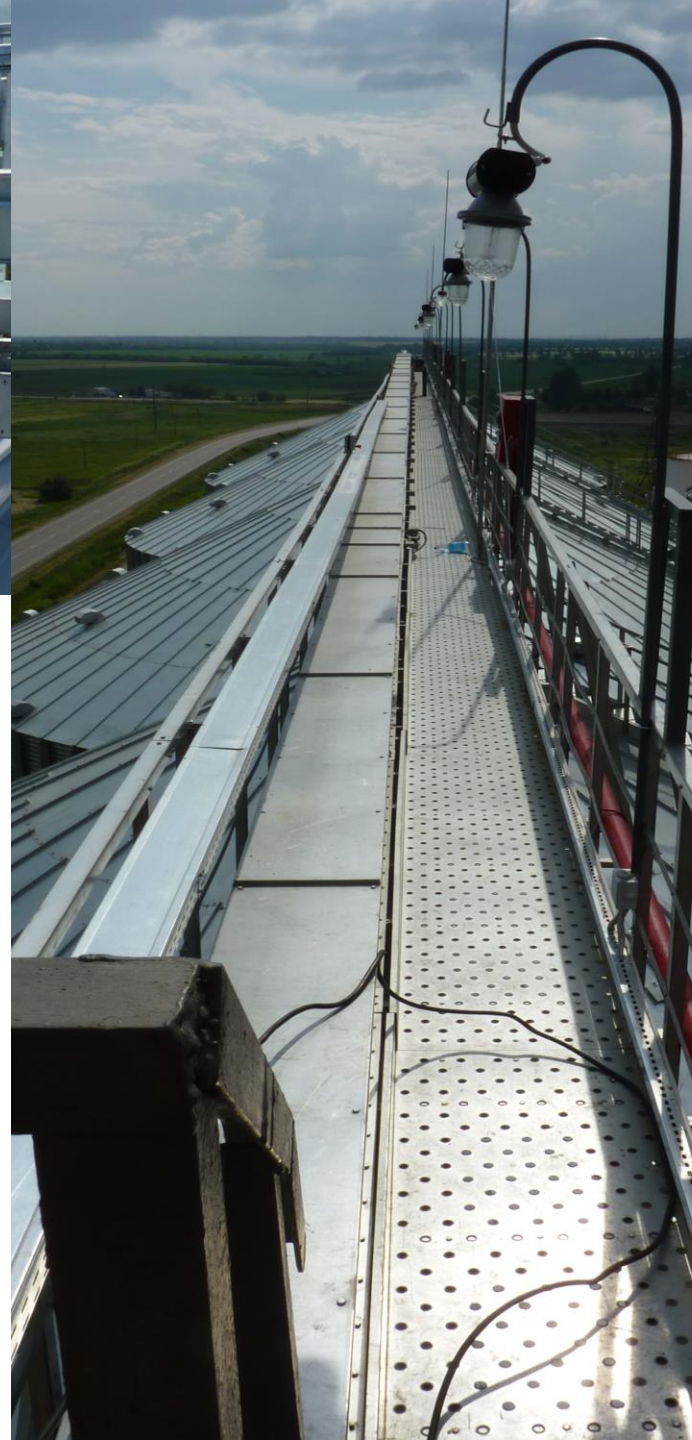
For fixing of carrying elements bolts 8,8 strength class are used. Fastening is manufactured according to DIN 933 norms; strength class 8.8 and 10.9 following to ISO 898-1:2009. Bolts have zinc coating made by hot galvanization according to EN-ISO 10684: 2006. All bolt joints are supplied with isolating gaskets, which prevent moisture intake through threaded connections.





Transport lines

Top transport galleries allow leaning on silo roof peak and fixing to strengthen vertical ribs without usage of separate carrying support. Maximum gap between support points (depending on silo diameter) can be not more than 15 meters, acceptable point vertical load – not more 1000 kg/linear meter on one gallery.



LADDERS

Vertical ladders with rails and rest platforms are bolted along the whole silo height. There is also roof ladder. It provide access to over silo gallery and thermal suspensions service hatches. All ladders are designed according to the security norms.





SILO UNLOADING

Silo is unloaded gravity flowing. Fully in conical hopper silos, partially in flat bottom bins. To remove grain remains screw sweeper conveyor in flat bottom silos is used.

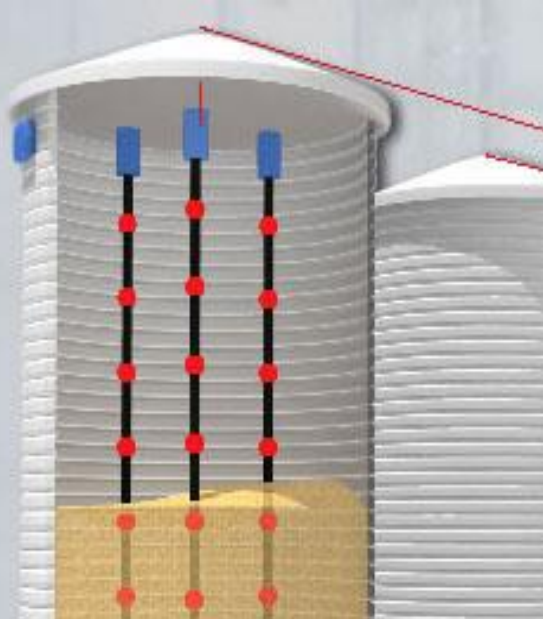
GRAIN ACTIVE VENTILATION

In conical bottom silo active ventilation system is located on the hopper sectors. Air distributors installed in it provide evenly passage of the air through grain column.

For flat bottom silos is used specially designed system of air ducts on aeration bottom which guarantee adequate ventilation.



STORAGE QUALITY



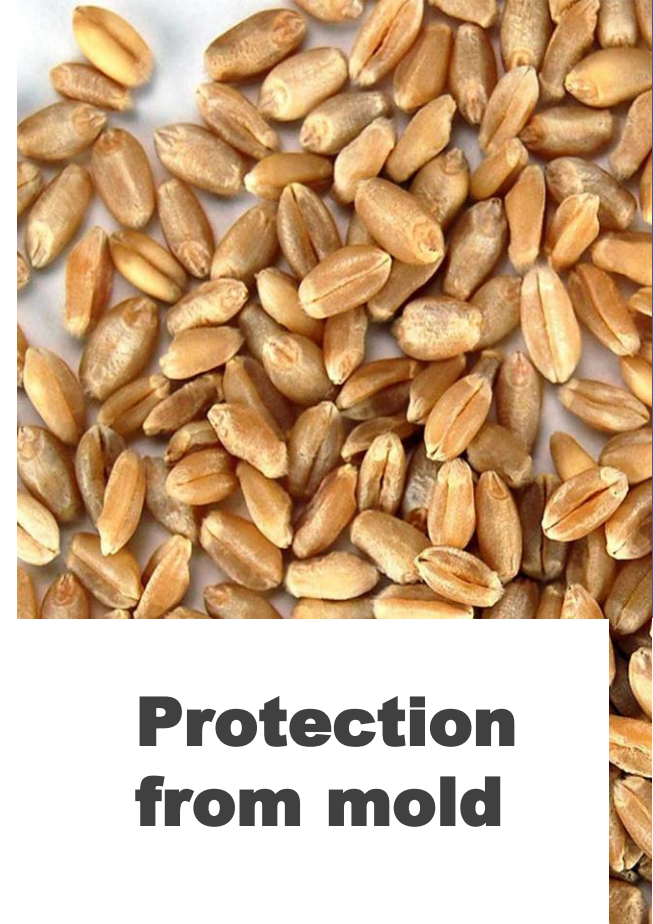
Temperature control

Cascade thermal suspension system.

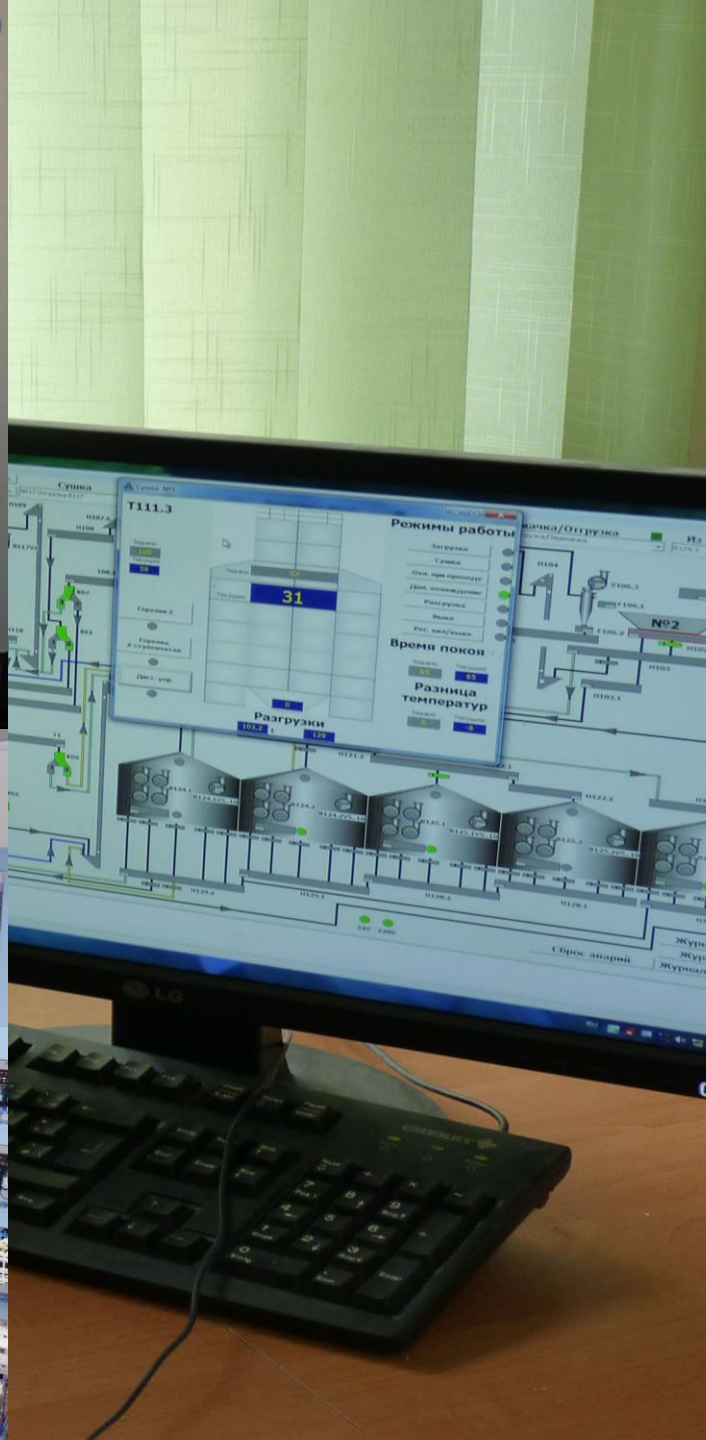
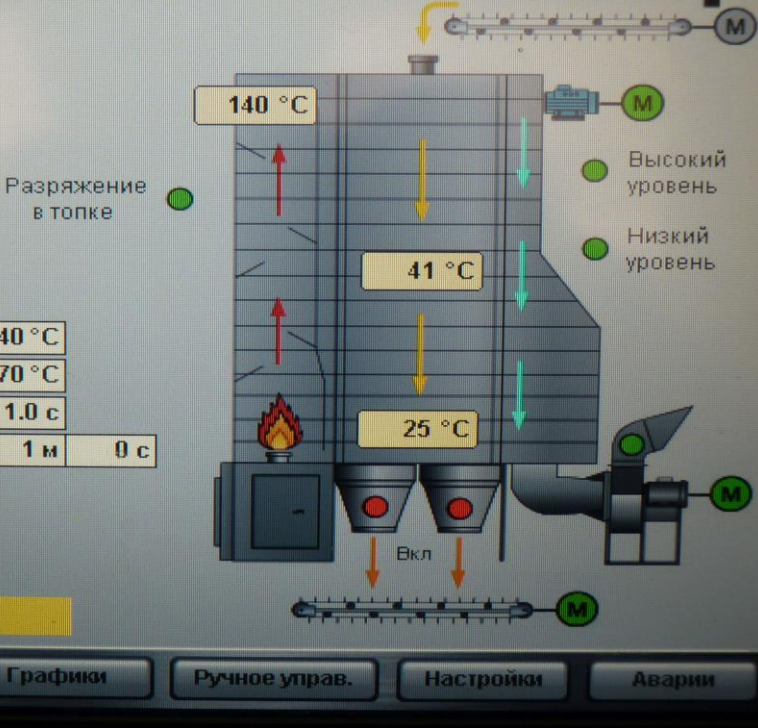
Thermal suspensions are equipped with sensors, hanging each 2 meters through the whole silo height.



Protection from rodents and insects



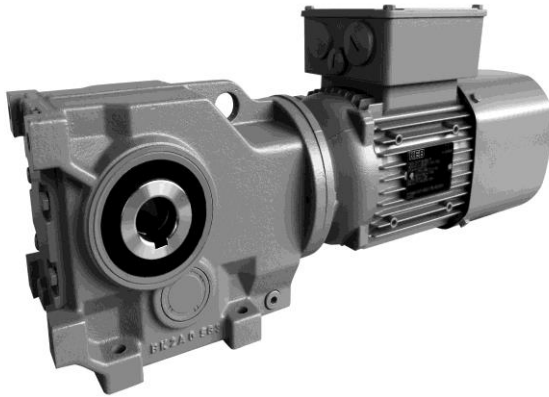
Protection from mold



Automatization of the management

Reduction of exploitation expenses in comparison with expenses for godown operation (for 1 storage ton of product). It's connected with possibility to provide full automatization of transport equipment, which decrease likelihood of human error.

Energy efficiency



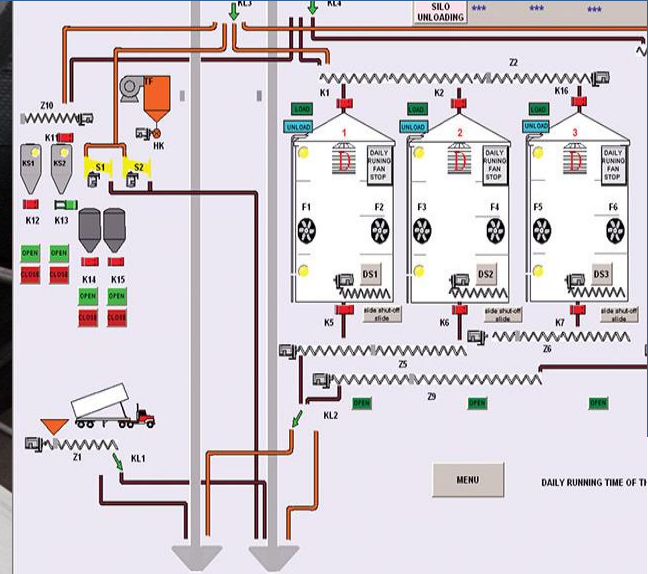
Modern effective drivers

Usage of motor-drivers with IE3 and IE4 energy efficiency class.



Economical transport equipment

Usage of self-made belt conveyors.



Optimization of transshipment routes

Optimally short technological routes with minimum amount of transport units.

LOW MOUNTAGE EXPENSES

Steel silos can be assembled as constructor from complete specially designed elements.





RECEPTION AND LOADING BINS

Typical modular solutions give an opportunity to fasten reception and loading operations in several times. It help to increase trucks circulation, trader doesn't pay for transport downtime.



TRANSPORT EQUIPMENT



LUBNYMASH manufactures the range of scrapper conveyors with capacity from 50 to 500 t/hour, belt conveyors from 25 to 550 t/hour, bucket elevators from 5 to 1000 t/hour, screw conveyors of different diameters and types.



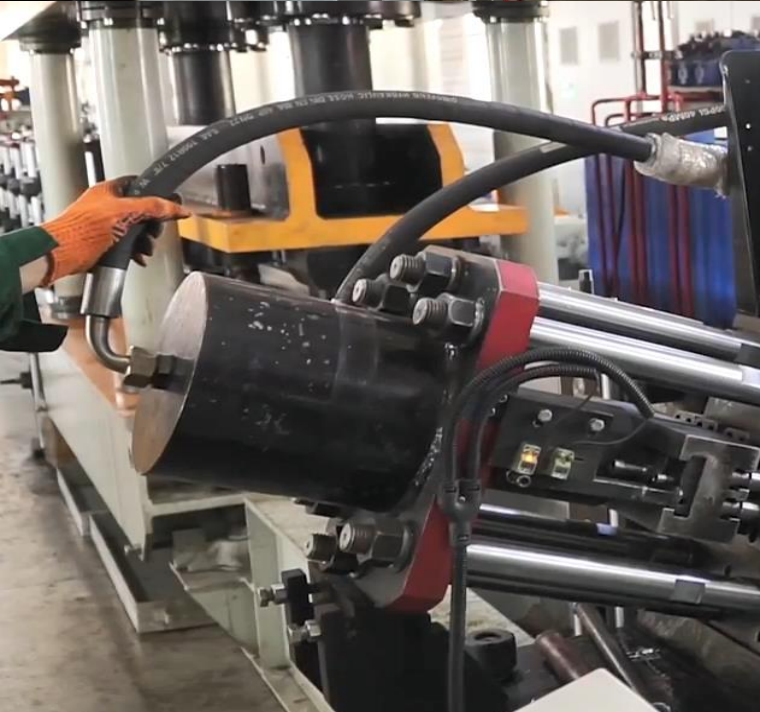
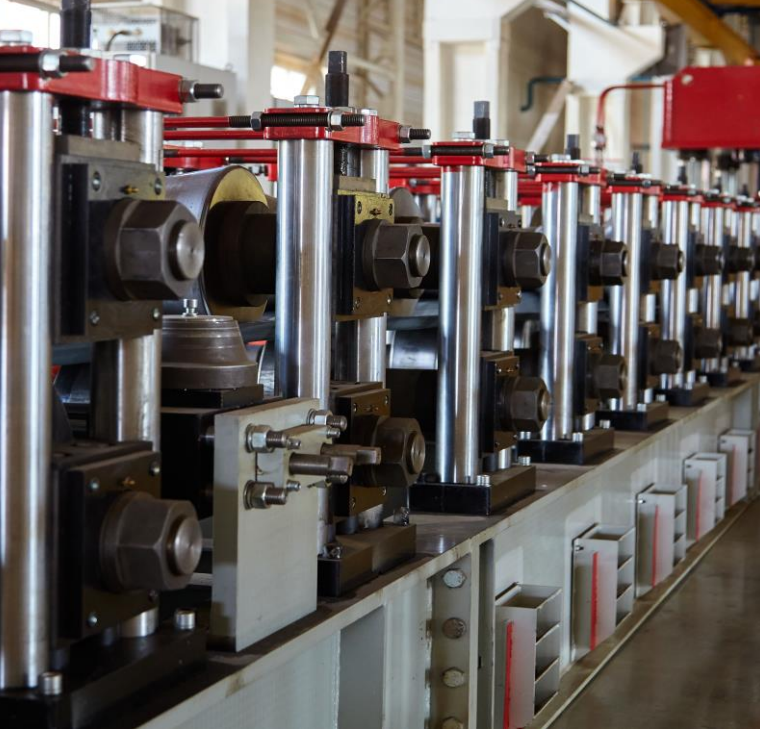
A large, multi-story industrial grain dryer with a corrugated metal exterior. It features several external ladders and walkways. Red signs with white text are visible on the structure, including "LUBN" and "LUBNYMASH". A red shield logo with a white stylized 'A' is also present.

MODERN DRYING EQUIPMENT

Dryer for different cereals and seeds give you an opportunity not to think much about product safety storage. Excessive moisture content (more then 14%) negatively effect on grain quality. More over, efficient usage of drying agent is very important for farmers and agrarians

MODERN PRODUCTION MACHINES

Silo details are produced on modern machines with maximum level of automatization, as a result the quality of units is under control.





OBJECTS MAP

LUBNYMASH has multiple long-term contracts with big enterprises from Ukraine, Belarus, Kazakhstan, Moldova and other countries.





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ISO 9001 : 2008
ISO 14001 : 2004
OHSAS
18001 : 2007
IS 18001 : 2007
SA 8000 : 2008
CE

